Aman Sharma

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.95
Manan Vidya Public School	Class XII (CBSE)	2018	75.00%
DAV Nandraj Public School	Class X (CBSE)	2016	8.40%

PROJECTS

Development of an Arc-Jet Plasma Facility for Material Erosion Testing (Affiliation: DRDO)

- Contributed to the design, setup, and commissioning of a 50 kW DC non-transferred arc-jet plasma facility
- Validated operational boundaries using thermocouples and IR thermal imaging techniques
- Tested candidate TPS materials under high-enthalpy plasma flow conditions
- Evaluated erosion behavior by measuring surface recession, mass loss, and surface morphology

MHD Modeling and Thermal Analysis of High-Enthalpy Arc-Jet Plasma Flows

- Developed a fully coupled 2D axisymmetric Magnetohydrodynamic (MHD) model
- Solved Maxwell's equations coupled with compressible Navier-Stokes and energy equations
- Evaluated arc dynamics for argon and air working gases at flow rates of 20 and 40 SLPM
- Predicted peak core plasma temperatures of 14,670 K and nozzle-exit temperatures of 4,700 K

Numerical Analysis of Material Ablation in High-Enthalpy Plasma Environments

- Built a 2D transient multiphysics model to simulate carbon-phenolic TPS ablation up to 300 W/cm² heat flux.
- Coupled solid-gas heat transfer, Arrhenius pyrolysis kinetics, and Darcy porous media flow.
- Tracked surface recession and moving boundaries using the Arbitrary Lagrangian-Eulerian (ALE) method
- Modeled the evolution of virgin, pyrolysis, and char layers under surface temperatures exceeding 2,700 K
- Validated model against published experimental data, achieving an agreement within 0.3% - 0.5% deviation

Deployment Mechanisms for CubeSat Solar Panels & Planetary Rover Doors

- Reduced solar panel vibration & hindrance during the CubeSat deployment phase by 55%.
- Developed two mechanisms: the first exploiting a spring hinge, and the second incorporating motorization.
- Delivered a presentation and project overview during the 4th Indian Planetary Science Conference (IPSC-2023) held at the Physical Research Laboratory, Ahmedabad.

Development & Mechanical Characterization of Hybrid Fiber Composites

- Fabricated four thermosetting composite configurations using epoxy resin (LY556/HY951): neat epoxy, coir-reinforced, mixed natural fiber (cane, pineapple, banana), and glass fiber-reinforced.
- Evaluated dynamic fracture resistance using standardized Izod and Charpy impact tests.
- Assessed structural shear strength and surface indentation behavior using torsion and Brinell hardness testing.
- Correlated fiber hybridization to achieve an optimized high-strength, low-density profile.

SKILLS

Libraries: Python, NumPy, Matplotlib, pandas

Software: COMSOL Multiphysics, Ansys Workbench, MATLAB, Fusion 360, Microsoft Office

Core Skills: MHD Modeling, Multiphysics Simulation, Thermal Analysis, Plasma Flow Analysis, Material Ablation

Soft Skills: Communication, Problem Solving, Teamwork, Leadership

HONORS/AWARD

- MIRAI Setu Fellow (2026): Selected for a fully funded Japan-India government exchange exploring Japanese mechanical manufacturing standards, automation, and advanced engineering.

- 9th SOF International Mathematics Olympiad: Awarded a gold medal for securing 1st rank at the school level in a competitive international mathematics examination.

INTERNSHIP

Numerical Investigation of Jet Impingement Cooling on Concave Surfaces | IIT Indore

- Performed CFD simulations of jet impingement cooling over concave curved surfaces to study heat-transfercharacteristics.
- Worked with near-wall mesh resolution (y+ 1) and transient simulation settings to accurately capture flow andthermal behavior near the surface.
- Analyzed temperature, velocity, and flow-field characteristics to study the effect of jet impingement on coolingperformance and assess numerical convergence

Feature on Wing Flap to Enhance Aerodynamic Efficiency | Simulation Lab

- Performed CFD-based aerodynamic analysis of a NACA 2414 wing with multi-element flaps, studying the influenceof flap configuration and deflection on aerodynamic performance.
- Conducted parametric simulations across different flap deflections and angles of attack to examine changes in lift,drag, and L/D characteristics.
- Interpreted pressure, velocity, lift, and drag results from CFD simulations to identify configurations with improvedsubsonic aerodynamic performance.

RELEVANT COURSES

Mathematical Methods for Engineers | Computational Fluid Dynamics | Tribological and surface engineering aspects of Lightweighting | Machine Learning and its Applications | High-speed Aerodynamics | Aerodynamics and Propulsion |

EXTRACURRICULAR

- Aksharadhaan Group - Volunteered in community service by teaching Class 10 government school students and supporting their academic development.

POSITIONS OF RESPONSIBILITY

Placement Coordinator - Coordinated campus placement activities, served as a liaison between students and recruiters, and assisted in organizing recruitment drives and placement events.

CONFERENCES AND PUBLICATIONS

- Numerical Investigation and Thermal Characterization of a Hydrogen Arc-Jet Plasma Facility for High-Enthalpy Re-Entry Applications--Accepted Oral Presentation, NAPC 2027, Jan. 2027.

- High-Speed Free-Flight Testing of Projectiles and Its Impact Response--Accepted Oral Presentation, NAPC2027, Jan. 2027.